

Table 1: Collision avoidance cost, cost\_limit=0: SafeDreamer (100k steps) vs MACPO (1000k steps).

| Algorithm     | Scenario     | Score $\uparrow$                   | Cost $\downarrow$                 | Winrate $\uparrow$                | Steps | Cost Limit |
|---------------|--------------|------------------------------------|-----------------------------------|-----------------------------------|-------|------------|
| MACPO         | 2s_vs_1sc    | 12.23 $\pm$ 0.48                   | 0.00 $\pm$ 0.00                   | 0.11 $\pm$ 0.04                   | 1000k | 0          |
| Safe Dreamers | 2s_vs_1sc    | <b>19.86 <math>\pm</math> 0.22</b> | 0.00 $\pm$ 0.00                   | <b>0.95 <math>\pm</math> 0.03</b> | 100k  | 0          |
| MACPO         | 3m           | 11.20 $\pm$ 0.96                   | 0.12 $\pm$ 0.03                   | 0.34 $\pm$ 0.07                   | 1000k | 0          |
| Safe Dreamers | 3m           | <b>19.03 <math>\pm</math> 0.91</b> | <b>0.02 <math>\pm</math> 0.02</b> | <b>0.92 <math>\pm</math> 0.07</b> | 100k  | 0          |
| MACPO         | 2s3z         | 9.24 $\pm$ 0.21                    | 0.07 $\pm$ 0.02                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 2s3z         | <b>18.72 <math>\pm</math> 0.44</b> | 0.16 $\pm$ 0.11                   | <b>0.79 <math>\pm</math> 0.07</b> | 100k  | 0          |
| MACPO         | 3s_vs_3z     | 8.32 $\pm$ 0.07                    | 0.00 $\pm$ 0.00                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 3s_vs_3z     | <b>19.89 <math>\pm</math> 0.05</b> | 0.02 $\pm$ 0.03                   | <b>0.97 <math>\pm</math> 0.01</b> | 100k  | 0          |
| MACPO         | 3s_vs_4z     | 5.77 $\pm$ 0.13                    | 0.00 $\pm$ 0.00                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 3s_vs_4z     | <b>15.49 <math>\pm</math> 3.71</b> | 0.26 $\pm$ 0.23                   | <b>0.06 <math>\pm</math> 0.09</b> | 100k  | 0          |
| MACPO         | 3s_vs_5z     | 4.78 $\pm$ 0.26                    | 0.00 $\pm$ 0.00                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 3s_vs_5z     | <b>13.72 <math>\pm</math> 2.97</b> | 0.08 $\pm$ 0.07                   | <b>0.01 <math>\pm</math> 0.03</b> | 100k  | 0          |
| MACPO         | 8m           | 7.21 $\pm$ 0.32                    | 0.70 $\pm$ 0.05                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 8m           | <b>17.49 <math>\pm</math> 0.43</b> | <b>0.58 <math>\pm</math> 0.06</b> | <b>0.71 <math>\pm</math> 0.05</b> | 100k  | 0          |
| MACPO         | MMM          | 6.92 $\pm$ 0.15                    | 1.50 $\pm$ 0.02                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | MMM          | <b>15.12 <math>\pm</math> 0.17</b> | <b>1.20 <math>\pm</math> 0.52</b> | <b>0.39 <math>\pm</math> 0.03</b> | 100k  | 0          |
| MACPO         | 3s5z_vs_3s6z | 6.83 $\pm$ 0.07                    | 0.10 $\pm$ 0.01                   | 0.00 $\pm$ 0.00                   | 1000k | 0          |
| Safe Dreamers | 3s5z_vs_3s6z | <b>10.65 <math>\pm</math> 0.82</b> | 0.86 $\pm$ 0.08                   | <b>0.01 <math>\pm</math> 0.01</b> | 100k  | 0          |
| MACPO         | bane_vs_bane | 19.66 $\pm$ 0.09                   | 0.65 $\pm$ 0.00                   | 0.88 $\pm$ 0.06                   | 1000k | 0          |
| Safe Dreamers | bane_vs_bane | 18.20 $\pm$ 0.59                   | <b>0.15 <math>\pm</math> 0.13</b> | 0.57 $\pm$ 0.10                   | 100k  | 0          |